

Alejandro Echeverria

Senior AI Engineer

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Professional Summary

A Senior AI Engineer with deep expertise in designing, building, and deploying AI-powered systems at scale. I architect end-to-end AI solutions—from data pipelines and model integration to production deployment and compliance—across cloud-native environments. I specialize in translating complex research and business requirements into secure, scalable AI products that drive measurable impact in global health, enterprise, and healthcare domains.

Technical Skills

AI & ML: Generative and analytical ML/AI, LLMs, AI agents, multi-agent systems, intelligent automation, design patterns; prompt engineering, prompt optimization, and token cost management (FinOps for AI); model selection and right-sizing strategies; Retrieval-Augmented Generation (RAG); latency optimization and performance benchmarking; model evaluation and A/B testing; Azure OpenAI; AI guardrails and Responsible AI practices.

Scalability & Performance: Distributed AI pipeline design; horizontal scaling with Kubernetes; streaming AI workloads (Event Hub); async/parallel pipeline architecture; caching strategies for AI services; Azure AI cost monitoring and budget governance.

Data & Cloud: Azure Data Factory, Synapse Analytics, Analysis Services, Data Lake Storage, Event Hub, Cosmos DB, Microsoft Fabric; SQL, NoSQL, Data Lake, Redis, Blob.

Languages: Python, C#, JavaScript, TypeScript, R, SQL, PowerShell.

Infrastructure & DevOps: Docker, Kubernetes; Azure DevOps, GitHub Actions, Drone; ARM, Terraform (IaC).

Architecture & Frameworks: Solutions Architecture; .NET, Node.js, Entity Framework; TCP/IP, HTTP, Zero Trust Network; SDL, Scrum, Agile, DevOps.

Security & Compliance: Threat modeling, SAML, OAuth2.0, SSO; GDPR, CCPA, HIPAA; Linux systems administration and internals.

Soft Skills

Delivery & Communication: Technical project management (scoping, coordinating AI workflows, budgeting, tracking deliverables); collaborative cross-team communication, both technical and non-technical; industry and research experience in global health and healthcare.

Analysis & Insight: Data-driven business insights; actionable metric creation from large raw data; attention to detail and problem-solving.

Professional: Leadership, initiative, motivation, and accountability; security-oriented, with a focus on data privacy and responsible use of AI; adaptability and continuous learning mindset.

Professional Experience

Senior Software Engineer (Data and AI) – Gates Foundation

2025 - Current

Seattle, WA

At the Institute for Disease Modeling (IDM), I design, build, and deploy AI and data systems that accelerate global health research. I architect end-to-end AI pipelines—from raw epidemiology and simulation data to interactive visualizations and decision-support tools—used by researchers and stakeholders to drive data-informed decisions in infectious disease eradication. I lead rapid prototyping and production delivery of AI solutions adopted by partner institutions and LMICs, identify gaps in research data infrastructure, and evaluate and recommend emerging AI technologies to maximize the Foundation's impact in Global Health.

Senior AI Solutions Consultant – Dura Digital

2025

Seattle, WA

As Senior AI Solutions Consultant, I led architectural guidance for enterprise Azure AI deployments. I designed a compliant Azure AI Foundry architecture for a regulated Canadian financial institution, encompassing multi-region deployment (Canada Central and Canada East) for data sovereignty under PIPEDA and BCFSA regulations, zero-trust security, private networking, MuleSoft Flex Gateway for Claude model routing, and compliance-grade observability with immutable audit logging. I evaluated model availability, identified architectural gaps, and provided effort estimates across a 13-task implementation roadmap. Dura Digital is a global IT consulting and services firm dedicated to delivering business, data, and AI-driven growth solutions.

Software Engineer II – Microsoft

2019 – 2025

Redmond, WA

I designed and shipped AI-powered enterprise products at scale, including AI Services, intelligent automation, speech recognition and analytics, and cognitive tools supporting millions of global users. I architected machine learning pipelines and integrated AI/ML capabilities into cloud-native applications using Azure infrastructure

and DevOps practices. I conducted threat modeling and implemented zero-trust strategies to secure AI services, and managed incident response for high-priority production incidents.

Software Engineer - Motiv Inc. (Microsoft Contract)

2017 – 2019

Redmond, WA

I developed secure, high-throughput cloud applications at Microsoft's Core Platform Engineering Group. I built APIs and microservices, identified and mitigated software vulnerabilities, enforced data privacy requirements, and implemented DevOps processes to meet enterprise-grade security and compliance standards.

Software Engineer - Getty Images

2012 – 2017

Seattle, WA

I developed and tested web services and databases handling confidential financial data and time-sensitive royalty calculation processes. Areas of experience included .NET development, relational databases, message brokers, monitoring and integration platforms, CI/CD, DevOps and Agile methodologies, and on-call engineering support. Getty Images is the global leader in multimedia and royalty services.

Software Engineer in Test - iSoftStone Inc.

2007 – 2012

Kirkland, WA

I designed and executed automated tests for web, mobile, and desktop software products. I developed unit and integration tests for web applications and APIs using automated frameworks and custom scripting. I led offshore and on-site testing teams for Fortune 100 technology clients.

Education

Bachelor of Science in Electronics Engineering

Universidad del Valle de Guatemala

Certifications

- Agile Project Management, Google
- Azure Security Engineer Associate, Microsoft
- Solutions Architecture, University of Washington
- Telecommunications and Networks, America Movil